



BAULKHAM HILLS HIGH SCHOOL

2015 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 11 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks) **Questions 1-10**

- Attempt Questions 1-10
Allow about 15 minutes for this section.

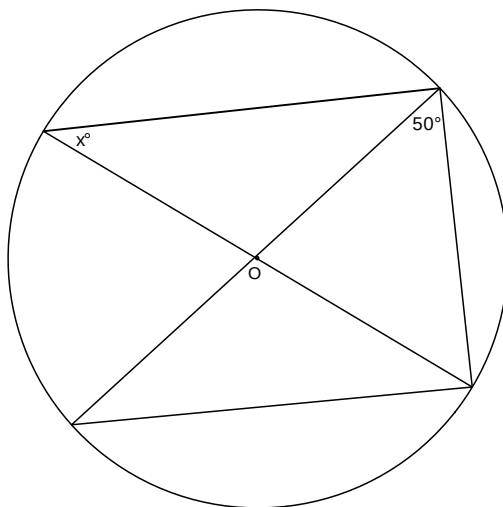
Section II – Pages 5-10 (60 marks)

- Attempt questions 11-14
Allow about 1 hour and 45 minutes for this section.

Table of Standard Integrals is on page 11

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. If O is the centre of the circle, the value of x in the following diagram is:

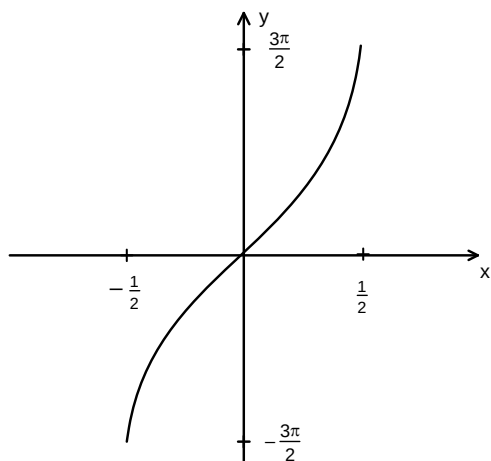


- (A) 25°
- (B) 40°
- (C) 50°
- (D) 80°
2. The point P divides the interval AB externally in the ratio 3 : 2. If A(-2,2) and B(8,-3) what is the y coordinate of the point P?
- (A) -13
- (B) -1
- (C) 4
- (D) 28
3. How many distinct arrangements of the letters of the word *ALGEBRA* are possible in a straight line if the A's are separated.
- (A) 720
- (B) 1800
- (C) 2160
- (D) 2520

4. The polynomial $P(x) = x^3 - 6x^2 - 2x + k$ has a factor of $x + 2$. What is the value of k ?
- (A) -28
(B) -20
(C) 20
(D) 28

5. The acute angle between the lines $4x + y = 2$ and $y = 2x - 1$ to the nearest degree is :
- (A) 12°
(B) 13°
(C) 40°
(D) 41°

6. The equation of an inverse trig function drawn below is :



- (A) $y = \frac{1}{3} \sin^{-1} \frac{x}{2}$
(B) $y = \frac{1}{3} \sin^{-1} 2x$
(C) $y = 3 \sin^{-1} \frac{x}{2}$
(D) $y = 3 \sin^{-1} 2x$
7. A particle moving in simple harmonic motion with displacement x and velocity v , is given by $v^2 = 9(16 - x^2)$. What is its amplitude (A) and its period (T) ?
- (A) $A=3$ $T = \frac{\pi}{2}$
(B) $A=3$ $T = \frac{2\pi}{3}$
(C) $A=4$ $T = \frac{\pi}{2}$
(D) $A=4$ $T = \frac{2\pi}{3}$

8. $\int \frac{1}{\sqrt{25-4x^2}} dx =$

(A) $\frac{1}{2} \sin^{-1} \frac{2x}{5} + c$

(B) $\frac{1}{2} \sin^{-1} \frac{4x}{25} + c$

(C) $\frac{1}{4} \sin^{-1} \frac{2x}{5} + c$

(D) $\frac{1}{4} \sin^{-1} \frac{4x}{25} + c$

9. The derivative of $\tan^{-1}x^4$ is:

(A) $\frac{1}{1+x^8}$

(B) $\frac{4x^3}{1+x^8}$

(C) $4(\tan^{-1}x)^3$

(D) $\frac{4(\tan^{-1}x)^3}{1+x^2}$

10. The solution to $|2x-1| \leq |x-2|$ is

(A) $x \leq -1$

(B) $x \geq 1$

(C) $-1 \leq x \leq 1$

(D) $x \leq -1$ or $x \geq 1$

End of Section 1

Section II – Extended Response**All necessary working should be shown in every question.**

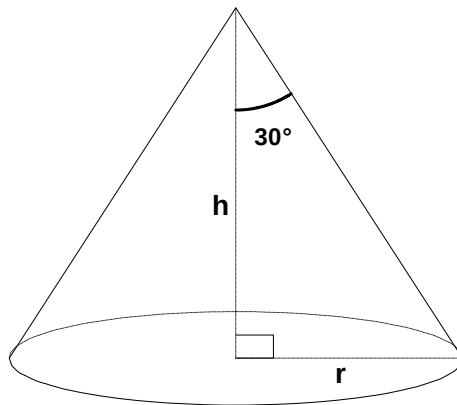
Question 11 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Solve $\frac{2}{3x-1} \leq 1$	3
b)	Find $\int \frac{2x \, dx}{(2x+1)^2}$ using the substitution $u = 2x + 1$	3
c)	Evaluate $\lim_{x \rightarrow 0} \frac{\sin \frac{1}{2}x}{3x}$	2
d)	Find the constant term in the expansion $\left(2x + \frac{3}{x^3}\right)^8$.	2
e)	(i) Show that a root of the continuous function $f(x) = x + \frac{1}{2}\sin 2x - \frac{\pi}{4}$ lies between 0.4 and 0.5.	1
	(ii) Hence use one application of Newton's method with an initial estimate of $x = 0.4$ to find a closer approximation for the root to 2 significant figures.	2
f)	Solve $\sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$	2
End of Question 11		

Question 12 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Evaluate $\int_0^{\frac{\pi}{8}} \sin^2 2x \, dx$	3
b)	(i) From a group of 6 boys and 6 girls, 8 are chosen at random to form a group. How many different groups of 8 people can be formed? (ii) How many of these groups consist of 4 boys and 4 girls? (iii) 4 boys and 4 girls are chosen and placed around a circle. What is the probability that the boys and girls alternate?	1 1 2
c)	The rate of change of the temperature (T) of an object is proportional to the difference between the temperature of the object and the temperature of the surrounding medium(C), ie $\frac{dT}{dt} = k(T - C)$ An object is heated and placed in a room of temperature $20^\circ C$ to cool. After 10 minutes its temperature is $36^\circ C$. After 20 minutes the temperature is $30^\circ C$. (i) Show $T = C + Ae^{kt}$ is a solution to the differential equation above. (ii) Find the value of A and the value of k to 3 decimal places. (iii) What was the temperature of the object when it was first placed in the room?	1 3 1
d)	Prove $(1)(3)(5) \dots (2n+1) = \frac{(2n+1)!}{2^n n!} \text{ for } n \geq 0 \text{ by mathematical induction.}$	3
End of Question 12		

Question 13 (15 marks) - Start on the appropriate page in your answer booklet	Marks
a) The polynomial $P(x) = x^3 - 6x^2 + kx - 8$ has roots α, β and γ. Find : (i) $\alpha + \beta + \gamma$ (ii) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$ (iii) k if $P(x)$ has a triple root.	1 2 2
b) PN is the normal to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$. The normal intersects the line SN which is parallel to the tangent at P. S is the focus of the parabola. (i) Show the equation of the normal PN is $x + py = 2ap + ap^3$. (ii) Find the equation of the line SN. (iii) Show that N has coordinates $(ap, ap^2 + a)$. (iv) Find the equation of the locus of N as P moves on the parabola.	2 1 2 2
Question 13 continues on the following page	

c) Sand is falling on the ground forming a conical pile whose semi apex angle is 30° .

The volume of the pile is increasing at a rate of $\frac{\pi}{100} m^3/s$. ($V = \frac{1}{3}\pi r^2 h$)



(i) Show that the volume of the pile is given by: $V = \frac{\pi h^3}{9}$

1

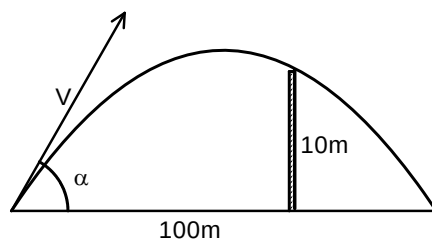
(ii) Find the rate at which the height of the pile is increasing when the height of the pile is 2 metres.

2

End of Question 13

Question 14 (15 marks) - Start on the appropriate page in your answer booklet

a)

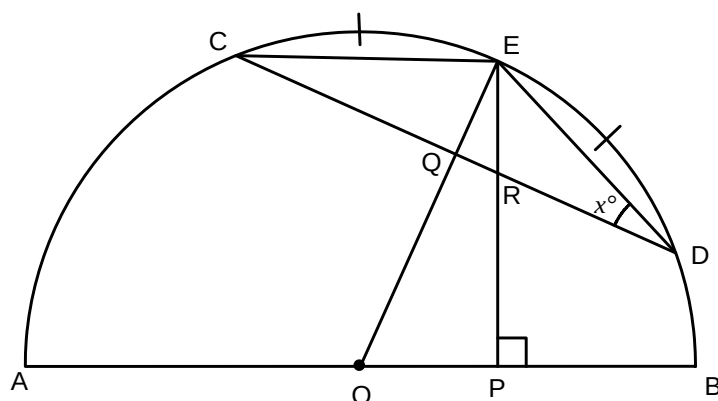


A projectile is fired from the ground with an angle of projection given by $\alpha = \tan^{-1} \frac{3}{4}$ and initial velocity V .

It just clears a wall $10m$ high $100m$ away. Let acceleration due to gravity be $g=10ms^{-2}$.

- | | | |
|-------|---|---|
| (i) | Show that the equations of motion are $x = \frac{4Vt}{5}$ and $y = -5t^2 + \frac{3Vt}{5}$. | 2 |
| (ii) | Find the initial velocity, V of the projectile. | 2 |
| (iii) | At what speed is the projectile travelling the instant it clears the wall? | 2 |

b) Copy or trace the diagram below in your exam booklet.

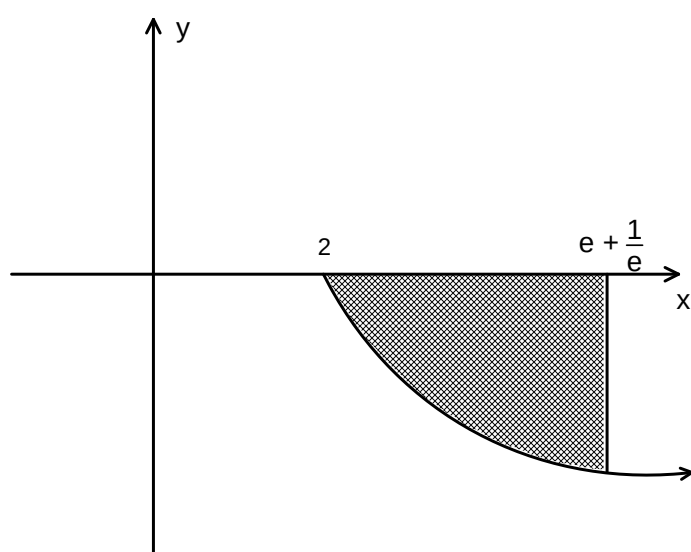


E is the midpoint of the arc CD . The radius OE meets CD at Q . EP is perpendicular to the diameter AB and meets CD at R .

- | | | |
|------|--|---|
| (i) | If $\angle CDE = x^\circ$, show $\angle EOD = 2x^\circ$. | 2 |
| (ii) | Prove $OPRQ$ is a cyclic quadrilateral. | 2 |

Question 14 continues on the following page

c) Below is the graph of $y = \ln \left(\frac{x - \sqrt{x^2 - 4}}{2} \right)$



(i) Show that the equation of the inverse function is given by $y = e^x + e^{-x}$.

2

(ii) Hence find the area of the shaded region above.

3

End of Paper.

M. Choice

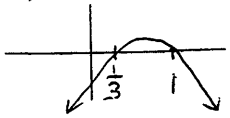
- 1) B 2) A 3) B 4) D 5) D
6) D 7) D 8) A 9) B 10) C

Question 11.

a) $\frac{2}{3x-1} \leq 1 \quad x \neq \frac{1}{3}$

$\frac{2(3x-1)^2 - 1(3x-1)^2}{(3x-1)^2} \leq 0 \quad \checkmark$

$(3x-1)(2-(3x-1)) \leq 0$
 $(3x-1)(3-3x) \leq 0 \quad \checkmark$



b) $\therefore \text{Soln } x < \frac{1}{3} \text{ or } x > 1$

$\int \frac{2x}{(2x+1)^2} dx \quad u = 2x+1$
 $\frac{du}{dx} = 2$

$\therefore \int \frac{2x}{(2x+1)^2} dx \quad dx = \frac{du}{2}$
 $2x = u-1$

$= \int \frac{u-1}{u^2} \cdot \frac{du}{2}$
 $= \frac{1}{2} \int \left(\frac{u}{u^2} - \frac{1}{u^2} \right) du$

2015 EXT.1. TRIAL SOLUTIONS: -

$= \frac{1}{2} \int \left(\frac{1}{u} - u^{-2} \right) du \quad \checkmark$
 $= \frac{1}{2} \left(\ln u + \frac{1}{u} \right) + C$
 $= \frac{1}{2} \left(\ln(2x+1) + \frac{1}{2x+1} \right) + C \quad \checkmark$

c) $\lim_{x \rightarrow 0} \frac{\sin \frac{1}{2}x}{3x}$
 $= \lim_{x \rightarrow 0} \frac{\sin \frac{1}{2}x}{\frac{1}{2}x} \cdot \frac{\frac{1}{2}x}{3x} \quad \checkmark$
 $= 1 \times \frac{1}{3} \quad \checkmark$
 $= \frac{1}{6} \quad \checkmark$

d) $(2x + \frac{3}{x^3})^8$
General term $= {}^8C_k (2x)^{8-k} (\frac{3}{x^3})^k$
 $= {}^8C_k 2^{8-k} x^{8-k} \frac{3^k}{x^{3k}} x^{-3k}$
 $= {}^8C_k 2^{8-k} 3^k x^{8-4k}$
 $\therefore 8-4k = 0$
 $k = 2$
 $\therefore \text{Constant term} = {}^8C_2 2^6 3^2$
 $= 16128 \quad \checkmark$

e) (i) $f(0.4) = -0.0267 \dots$

$f(0.5) = 0.135 \dots$

since $f(0.4) < 0$ & $f(0.5) > 0 \quad \checkmark$
a root exists between 0.4 & 0.5

(ii) $f'(x) = 1 + \cos 2x$

$\therefore a_1 = 0.4 - \left(\frac{-0.0267}{1.696} \right)$
 $= 0.4157 \dots \quad \checkmark$
 $a_1 = 0.42 \quad \checkmark$

f) $\sin 2\theta = \cos \theta$
 $2 \sin \theta \cos \theta - \cos \theta = 0$
 $\cos \theta (2 \sin \theta - 1) = 0$
 $\sin \theta = \frac{1}{2} \quad \cos \theta = 0$
 $\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}, \frac{3\pi}{2}$

Question 12

a) $\int_0^{\frac{\pi}{8}} \sin^2 2x \, dx$

$= \frac{1}{2} \int_0^{\frac{\pi}{8}} (1 - \cos 4x) \, dx \quad \checkmark$

$= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{8}} \quad \checkmark$

$= \frac{1}{2} \left[\left(\frac{\pi}{8} - \frac{\sin \frac{\pi}{2}}{4} \right) - (0 - 0) \right]$

$= \frac{1}{2} \left[\frac{\pi}{8} - \frac{1}{4} \right]$
 $= \frac{\pi}{16} - \frac{1}{8} \text{ or } \frac{\pi-2}{16} \quad \checkmark$
 $= (0.0713 \dots) \quad \checkmark$

b) (i) ${}_{12}C_8 = 495 \quad \checkmark$

(ii) ${}_6C_4 \times {}_6C_4 = 225 \quad \checkmark$

(iii) $\frac{\text{Alternating}}{\text{Total}} = \frac{4! \times 3!}{7!} \quad \checkmark$
 $= \frac{1}{35}$
(or equivalent)

c) (i) $T = C + Ae^{kt} \rightarrow Ae^{kt} = T - C \quad \text{--- (1)}$

$\frac{dT}{dt} = kAe^{kt}$ from (1) $\checkmark$
 $= k(T - C)$

$\therefore T = C + Ae^{kt}$ is a soln

Given when $t = 10$ $T = 36$ $C = 20$

$\therefore 36 = 20 + Ae^{10k}$

$Ae^{10k} = 16 \quad \text{--- (1)}$

when $t = 20$ $T = 30$

$\therefore 30 = 20 + Ae^{20k}$

$\therefore 10 = Ae^{20k} \quad \text{--- (2)}$

(2) $\div$ (1) $\frac{10}{16} = \frac{Ae^{20k}}{Ae^{10k}}$

$\frac{5}{8} = e^{10k}$

$\therefore k = \frac{\ln(\frac{5}{8})}{10} = (-0.047) \quad \checkmark$

Sub into (2) $10 = Ae^{-0.047(20)}$

$A = 25.6 \quad \checkmark$

$T = 20 + 25.6e^{kt}$

when $t = 0$

$T = 20 + 25.6e^0$
 $= 45.6 \quad \checkmark$

d) $(1)(3)(5) \dots (2n+1) = \frac{(2n+1)!}{2^n n!} \quad n \geq 0$

Step 1.
Prove true for $n=0$
LHS $\therefore 2(0)+1 = 1$ RHS $= \frac{(2(0)+1)!}{2^0 0!}$
 $= \frac{1}{1} = 1$

Step 2: LHS = RHS (true for $n=0$)

Assume true for $n=k$ i.e.

(1)(3)(5) ... $(2k+1) = \frac{(2k+1)!}{2^k k!}$

Step 3
Prove true for $n=k+1$ i.e.

(1)(3) ... $(2k+1)(2k+3) = \frac{(2k+3)!}{2^{k+1} (k+1)!} \quad \checkmark$

i.e. $\frac{(2k+1)!}{2^k k!} \cdot (2k+3) = \frac{(2k+3)!}{2^{k+1} (k+1)!}$

$= \frac{(2k+1)! \cdot (2k+2)(2k+3)}{2^k k! (2k+2)}$

$= \frac{(2k+3)!}{2^{k+1} (k+1)!} \quad \checkmark$

$= \frac{(2k+3)!}{2^{k+1} (k+1)!}$

Step 4 = RHS.

$\therefore$ True for $n=k+1$ if true for $n=k$. Therefore true for $n=0, n=1, n=2 \dots$ & for all n by M. Induction.

a) $P(x) = x^3 - 6x^2 + kx - 8$

(i) $x + p + r = 6$ ✓
 (ii) $x^2p + x^2r + xpr^2$
 $= xpr(x + p + r)$ ✓
 $= 8(6)$ ✓
 $= 48$ ✓

(iii) If $P(x)$ has a triple root then
 $x^3 = 8 \rightarrow x = 2$ ✓
 sum of roots 2 at a time
 $3x^2 = k$
 $\therefore 3(2)^2 = k$
 $k = 12$ ✓

b) (i) $y = \frac{x^2}{4a}$
 $y' = \frac{2x}{4a}$
 at $x = 2ap$ $y' = \frac{2ap}{2a} = p$ ✓
 $\therefore$ Normal
 $y - ap^2 = -\frac{1}{p}(x - 2ap)$ ✓
 $py - ap^3 = -x + 2ap$
 $x + py = 2ap + ap^3$ ①

(ii) grad. of SP = p
 $\therefore$ SN $\rightarrow y = px + a$ ✓ - (2)

(iii) $y = px + a$ sub into ①
 $x + p(px + a) = 2ap + ap^3$
 $x + p^2x + ap = 2ap + ap^3$
 $x(1 + p^2) = ap^3 + ap$
 $x = \frac{ap(p^2 + 1)}{(1 + p^2)}$
 $x = ap$ sub into ② ✓
 $y = ap^2 + a$ ✓
 $\therefore N(ap, ap^2 + a)$

(iv) $x = ap$ $y = ap^2 + a$
 $\checkmark p = \frac{x}{a} \rightarrow y = a\left(\frac{x^2}{a^2}\right) + a$
 $y = \frac{x^2}{a} + a$ ✓
 $x^2 = ay - a^2$
 $x^2 = a(y - a)$

$V = \frac{1}{3} \pi r^2 h$
 $r = h \tan 30^\circ$
 $r = \frac{h}{\sqrt{3}}$ sub in

$\therefore V = \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}}\right)^2 h$
 $= \frac{1}{3} \pi \frac{h^3}{3}$
 $V = \frac{\pi h^3}{9}$ ✓

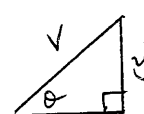
(ii) find $\frac{dh}{dt}$ when $h = 2$

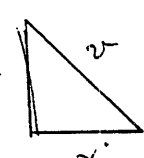
$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

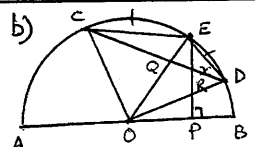
$\frac{dV}{dh} = \frac{2\pi h^2}{3}$ ✓

$\frac{\pi}{100} = \frac{2\pi(2)}{3} \cdot \frac{dh}{dt}$

$\frac{dh}{dt} = \frac{\pi}{100} \times \frac{3}{4\pi}$ ✓
 $= \frac{3}{400} \text{ m/s}$

14a) $\tan \theta = \frac{3}{4} \therefore \sin \theta = \frac{3}{5}$
 $\cos \theta = \frac{4}{5}$

 $\ddot{x} = 0$
 $\dot{x} = c_1$
 when $t = 0$ $y' = V \sin \theta$
 $\dot{x} = V \cos \theta$
 $\therefore \dot{x} = \frac{4V}{5}$ ✓
 $x = \frac{4tV}{5}$
 $y' = -10$
 $y' = -10t + c_2$
 when $t = 0$ $y' = V \cdot \frac{3}{5}$
 $\therefore y' = -10t + \frac{3V}{5}$ ✓
 $y = -5t^2 + \frac{3tV}{5} + c_3$
 when $t = 0$ $y = 0 \Rightarrow c_3 = 0$
 $\therefore y = -5t^2 + \frac{3tV}{5}$
 when $x = 100$ $y = 10$
 $\therefore 100 = \frac{4tV}{5}$ $10 = -5t^2 + \frac{3tV}{5}$
 $tV = 125$ ②
 sub ② into ①

$10 = -5t^2 + \frac{3(125)}{5}$
 $t = \sqrt{13}$ ✓
 sub into ②
 $V\sqrt{13} = 125$
 $= \frac{125}{\sqrt{13}}$
 $= 34.67$ ✓
 at $(100, 10)$ 
 need y' & x' ✓
 at $t = \sqrt{13}$
 $\dot{x} = \frac{4(34.67)}{5}$
 $= 27.73 \dots$
 $y' = -10\sqrt{13} + \frac{3}{5}(34.67)$
 $= -15.25 \dots$
 $v = \sqrt{(27.73)^2 + (-15.25)^2}$
 $v = 31.64 \dots$ ①

b)  Construct OD & OC

(i) $\angle EDC = x^\circ$ (Given)
 $\checkmark \angle CED = x^\circ$ (Equal chord subtend $\angle$'s at the circumference)
 $\therefore \angle EOD = 2x^\circ$ ($\angle$ at centre = 2 times $\angle$'s at circumference on same arc)

(ii) $OE = OD$ (Radii)
 $\angle OED = \angle ODE$ (opp. $\angle$'s in Δ)
 $\therefore \angle ODE = \frac{180 - 2x}{2} = 90 - x$
 $\angle EDC + \angle ODC = 90 - x$
 $x + \angle ODC = 90 - x$
 $\angle ODC = 90 - 2x$ ✓
 $\angle OAD + 90 - 2x + 2x = 180^\circ$ ✓
 ($\angle$ sum ΔOAD)
 $\therefore \angle OAD = 90^\circ$
 $\angle OPE = 90^\circ$ ($EP \perp OB$)
 $\therefore \angle OAR = \angle OPE = 90^\circ$ ✓
 $\therefore ORPO$ is a cyclic quad.
 (opposite angles of a cyclic quad. are supplementary)

$$(14c) \quad y = \ln\left(\frac{x - \sqrt{x^2 - 4}}{2}\right)$$

inverse
✓ $x = \ln\left(\frac{y - \sqrt{y^2 - 4}}{2}\right)$

$$e^x = \frac{y - \sqrt{y^2 - 4}}{2}$$

$$2e^x = y - \sqrt{y^2 - 4}$$

$$\sqrt{y^2 - 4} = y - 2e^x \quad \checkmark$$

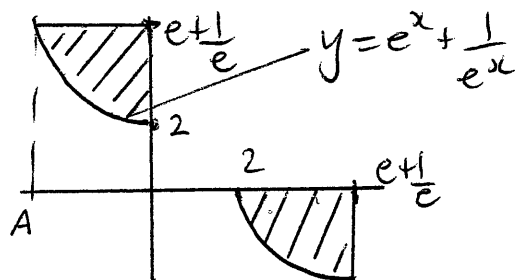
$$y^2 - 4 = y^2 - 4ye^x + 4e^{2x}$$

$$4ye^x = 4e^{2x} + 4$$

$$y = \frac{4e^{2x}}{4e^x} + \frac{4}{4e^x}$$

$$y = e^x + \frac{1}{e^x}$$

$$\underline{y = e^x + e^{-x}} \quad \checkmark$$



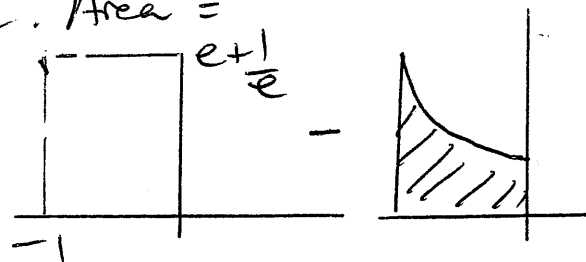
2 areas are equal because of symmetry.

At A $y = e + \frac{1}{e}$

$$\therefore e + \frac{1}{e} = e^x + \frac{1}{e^x}$$

✓ $x = \pm 1$ by inspection.
here $x = -1$

$\therefore$ Area =



$$\begin{aligned} &= \left(e + \frac{1}{e}\right) \times 1 - \int_{-1}^0 e^x + e^{-x} \quad \checkmark \\ &= e + \frac{1}{e} - (e^x - e^{-x})_{-1}^0 \\ &= e + \frac{1}{e} - (e^0 - e^0 - (e^{-1} - e^1)) \\ &= e + \frac{1}{e} - (1 - 1 - \frac{1}{e} + e) \end{aligned}$$

$$\begin{aligned} &= e + \frac{1}{e} + \frac{1}{e} - e \\ &= \frac{2}{e} \quad \checkmark \end{aligned}$$

NB AWARD 2 for

$$\int_{-1}^0 e^x + e^{-x} = e - \frac{1}{e}$$

AWARD 2

$$\int_0^1 e^x + e^{-x} = e - \frac{1}{e}$$